

For this use case, I would make the analyzer **measurement-driven rather than using arbitrary “AI quality scores.”** One important correction to the earlier chart: there is no universal numeric “ideal” for things like sharpness, noise, contrast, or bitrate. Their acceptable range depends heavily on scene content. ITU-T P.910 explicitly treats spatial information (SI) and temporal information (TI) as important scene characteristics, and current objective metrics such as VMAF combine several perceptual features rather than relying on one number.  ITU +1

So below is a practical **100-point scoring system for real-world journalism footage**—press conferences, speeches, demonstrations, interviews, mobile live streams, etc.

1. Master 100-point system

Parameter	Weight	Primary measurable metric	Direction
Compression artifacts	15	Blockiness + ringing + CAMBI	Lower artifact = better
Blur / motion blur	15	Variance of Laplacian + edge width + motion-aware blur	Less blur = better
Temporal consistency	15	Temporal flicker + optical-flow consistency	More stable = better
Texture / detail recovery	15	VIF / ADM / local high-frequency energy	More <i>credible</i> detail = better
Sharpness	10	MTF50 / Laplacian variance / edge strength	Optimum range
Noise	8	MAD / wavelet noise estimate	Less unwanted noise = better
Resolution / spatial detail	7	Effective resolution / MTF50 / edge density	Higher effective detail = better
Exposure / dynamic range	6	Clipping ratio + histogram statistics	Less clipping = better
Color / white balance	5	Gray-world error + ΔE when reference exists	Lower error = better
Stabilization	4	Global motion / residual camera motion	Less unwanted motion = better
Total	100		

Important: VMAF itself should *not* simply become one of these ten scores. It is better used as an additional **overall perceptual QA metric**, especially when you have a reference video. VMAF incorporates features such as VIF, ADM and motion, and its current implementation also supports PSNR, SSIM, MS-SSIM and CIEDE2000.  [GitHub](#) +1

2. Compression Artifact — 15 points

This is probably your **most important metric** because you're dealing with downloaded/re-uploaded low-bitrate live footage.

Measure

Use several indicators:

A. Blockiness

For an 8×8 block grid:

$$B = \frac{1}{N} \sum |I(x, y) - I(x + 1, y)|$$

at block boundaries, compared with neighboring non-boundary edges.

A more useful normalized version:

$$B_{norm} = \frac{E_{boundary}}{E_{nonboundary} + \epsilon}$$

High abnormal boundary energy → likely blocking.

B. Ringing

Measure oscillations around strong edges.

For an edge profile:

$$R = \frac{\text{secondary edge energy}}{\text{primary edge energy}}$$

C. CAMBI

CAMBI is particularly useful for detecting **banding**, which can appear after aggressive compression. It is available as a libvmaf feature.  [GitHub](#)

Practical thresholds

Artifact condition	Score
Virtually invisible	90–100
Mild	75–89
Noticeable at 100%	55–74
Clearly visible	30–54
Severe blocky/mosaic appearance	0–29

Processing trigger

`0–20 artifact severity → Skip
20–40 → Light artifact reduction
40–60 → Moderate restoration
60–80 → Strong restoration
80–100 → Heavy restoration, but protect texture`



Do not automatically denoise heavily. Compression noise and genuine skin texture can look similar.

3. Blur / Motion Blur — 15 points

This needs more than one metric.

A. Variance of Laplacian

For grayscale image I :

$$L = \text{Var}(\nabla^2 I)$$

Higher value generally means stronger high-frequency detail.

But this is **content-dependent**. A plain wall can have a low score even when perfectly sharp.

Therefore, calculate it only on:

- faces
- text
- edges
- high-gradient regions

Example normalized thresholds

For a standardized 1080p analysis crop:

Laplacian/edge score	Interpretation
> 300	Very sharp
150–300	Good
80–150	Soft
40–80	Blurry
< 40	Severe blur

These numbers are **engineering starting points**, not **universal standards**. You should calibrate them using your own newsroom footage.

B. Edge width

Take an edge profile and measure its 10–90% transition width:

$$W_{10-90} = x_{90} - x_{10}$$

Smaller width → sharper edge.

For example:

1–2 px = very sharp
2–3 px = good
3–5 px = soft
>5 px = significantly blurred



This is often more useful than Laplacian variance alone.

4. Temporal Consistency — 15 points

This is extremely important for your project.

You don't want:

Frame 1 → beautiful
Frame 2 → beautiful
Frame 3 → different texture
Frame 4 → different texture



That creates AI flicker.

A. Temporal flicker

Calculate aligned frame difference:

$$F_t = \frac{1}{N} \sum |I_t(x, y) - W(I_{t-1})(x, y)|$$

where W is optical-flow warping.

Then:

$$F_{temporal} = 1 - \text{normalized}(F_t)$$

Higher = more stable.

Suggested classification

Temporal consistency	Meaning	
95–100	Excellent	
90–94	Very good	
85–89	Good	
75–84	Noticeable instability	
<75	Problematic	

Again, these are **pipeline thresholds**, not an industry universal.

Even better

Use:

- optical flow
- forward-backward consistency
- temporal SSIM
- motion-compensated difference

rather than raw frame difference.

VMAF itself includes a motion feature based on temporal differences between adjacent frames.



5. Texture / Detail — 15 points

This is where many AI enhancers fail.

You don't want:

more texture

You want:

credible texture that remains temporally stable.

Use:

VIF

Visual Information Fidelity:

$$VIF = \frac{\text{information preserved}}{\text{information available}}$$

VIF is one of the core VMAF features. [GitHub](#)

ADM

Detail-loss measurement:

$$ADM \rightarrow \text{structural/detail preservation}$$

Also a VMAF core feature. [GitHub](#)

Local high-frequency ratio

$$HF = \frac{E_{high-frequency}}{E_{total}}$$

But do not simply maximize HF.

Oversharpening also increases high-frequency energy.

Therefore:

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Detail ↑

+

Temporal consistency ↑

+

Edge overshoot ↓

should determine whether detail is actually legitimate.

Texture score

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Natural texture	90-100
Good	75-89
Slightly mushy	55-74
Very soft	30-54
Texture destroyed	0-29

6. Sharpness — 10 points

This is deliberately separate from blur.

You need an **optimal zone**.

For example:

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Too soft ← GOOD → Oversharpened
 ▲
 target

Measure:

MTF50

If you have a suitable edge:

$$MTF(f) = \frac{FFT(LSF)}{FFT(LSF)(0)}$$

Then determine the spatial frequency where:

$$MTF = 0.5$$

That's **MTF50**.

Higher MTF50 generally means better resolving ability.

But:

Higher ≠ always better.

Artificial sharpening can produce halos while increasing edge strength.

Halo measurement

Measure overshoot/undershoot around edges:

$$H = \frac{E_{overshoot} + E_{undershoot}}{E_{edge}}$$

Target:

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<2%	Excellent
2-5%	Acceptable
5-10%	Visible
>10%	Oversharpened

These should be treated as **engineering thresholds**, not absolute perceptual standards.

7. Noise — 8 points

Use robust noise estimation rather than simple standard deviation.

One useful estimator:

$$\sigma \approx 1.4826 \times MAD$$

where

$$MAD = median(|x - median(x)|)$$

Better still, estimate noise in relatively flat regions.

Example normalized target

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Noise level	Score
Very low / clean	90-100
Low	75-89
Moderate	55-74
High	30-54
Severe	0-29

But don't make:

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noise = 0



your target.

Real camera footage naturally contains sensor texture, and aggressive denoising can destroy:

- skin pores
- hair
- fabric
- paper texture

8. Resolution / Spatial Detail — 7 points

Don't score resolution purely as:

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1920 × 1080 = 1080p = good

That's inadequate.

Measure:

Effective resolution

Use edge/MTF analysis.

For example:

$$ER = \frac{MTF50_{measured}}{MTF50_{reference}}$$

if you have a suitable reference.

Without reference:

- edge transition width
- MTF50
- local detail density
- text readability

are better indicators.

Practical interpretation

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90-100 → detail close to native

75-89 → good

55-74 → moderate loss

30-54 → substantial loss

0-29 → heavily degraded

And this is why a 1080p low-bitrate live stream can look worse than a good 720p source.

9. Exposure / Dynamic Range — 6 points

This can be measured very objectively.

Calculate:

$$C_{black} = \frac{\text{pixels} < T_{black}}{N}$$

and

$$C_{white} = \frac{\text{pixels} > T_{white}}{N}$$

For normalized 8-bit video, practical starting thresholds could be:

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Black clipping: $Y < 5$

White clipping: $Y > 250$

Then:

$$C_{clip} = C_{black} + C_{white}$$

Interpretation

Clipping	Condition
<1%	Excellent
1–3%	Usually acceptable
3–5%	Noticeable
5–10%	Problem
>10%	Severe

But scene-aware masking is essential.

A white sky shouldn't automatically be classified as exposure failure.

10. Color / White Balance — 5 points

For journalism footage, don't attempt cinematic grading automatically.

A. White balance

Estimate neutral regions:

$$WB_{error} = \sqrt{(R - G)^2 + (B - G)^2}$$

Lower = more neutral.

B. If reference exists: CIEDE2000

$$\Delta E_{00}$$

This is much better for color difference than raw RGB distance.

CIEDE2000 is available among libvmaf's additional features. [GitHub](#)

Practical interpretation

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```
 $\Delta E_{00} < 2$     → very close  
2–5              → small visible difference  
5–10            → noticeable  
>10             → strong color difference
```

These values are commonly useful engineering bands, but actual perceptual acceptability depends on viewing conditions and content.

11. Stabilization — 4 points

Estimate global camera motion using:

- feature matching
- optical flow
- affine/homography estimation

Let camera transform be:

$$H_t$$

Then measure:

$$M_t = \|H_t - H_{t-1}\|$$

After stabilization, residual global motion should decrease.

Score

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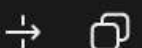


Residual unwanted motion	
< threshold	→ 90–100
low	→ 75–89
moderate	→ 55–74
high	→ 30–54
severe	→ 0–29

But don't stabilize intentional camera movement.

For example:

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camera operator pans toward speaker

is not necessarily a defect.

12. Overall score formula

After converting every parameter to 0–100:

$$Q = 0.15C + 0.15B + 0.15T + 0.15D + 0.10S + 0.08N + 0.07R + 0.06E + 0.05Color + 0.05Stab$$

where:

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C = Compression
B = Blur
T = Temporal
D = Detail/Texture
S = Sharpness
N = Noise
R = Resolution
E = Exposure
Color = Color
Stab = Stabilization

13. But I would NOT use only one overall score

For your software, I'd produce **three different scores**.

A. Visual Quality Score

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0–49 Poor
50–64 Fair
65–74 Good
75–84 Very good
85–94 Excellent
95–100 Reference-level / exceptionally clean

B. Restoration Need Score

This is more useful for your pipeline.

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+

→

□

Compression	78
Blur	64
Temporal	21
Noise	35
Color	12
Stabilization	48

This tells the processing engine **what to do**.

C. Artifact Risk Score

This prevents AI from making the video worse.

For example:

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→

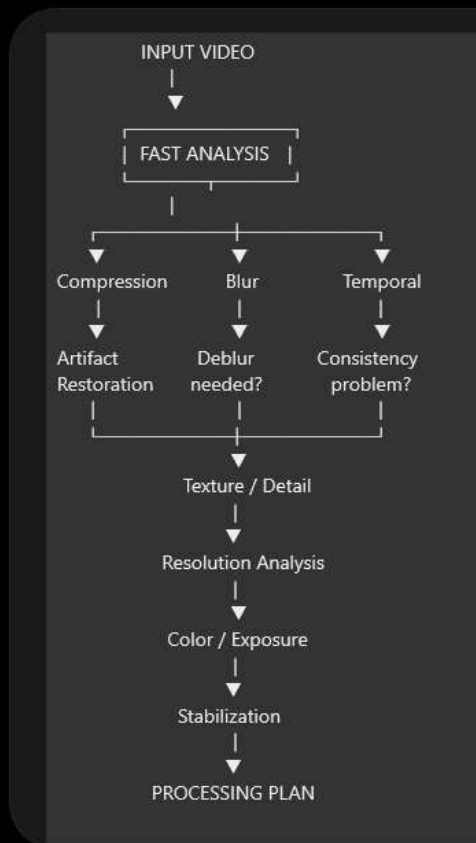
□

AI hallucination risk	22
Oversharpening risk	31
Temporal flicker risk	18
Plastic-skin risk	14

That is particularly important for your real-human footage.

14. The actual decision engine

This is the part I'd consider more important than the 100-point score.



Then something like:

```
IF compression > 40:
    artifact_restoration = ON

IF blur > 35:
    deblur = ON

IF temporal < 85:
    temporal_restoration = ON

IF texture < 60:
    restoration/VSR = ON

IF noise < 25:
    denoise = OFF

IF color > 85:
    color_correction = OFF

IF stabilization > 70:
    stabilization = ON
```

The actual thresholds should be calibrated against your footage rather than treated as universal.

15. One more metric I strongly recommend: VMAF

For before vs after testing, calculate:

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```
VMAF
VMAF temporal behavior
SSIM
MS-SSIM
PSNR
CAMBI
VIF
ADM
```

VMAF already exposes these kinds of underlying measurements, including VIF, ADM, motion, CAMBI, CIEDE2000, MS-SSIM, PSNR and SSIM depending on the build. [GitHub](#)

However, there's an important limitation:

VMAF needs a reference video for its conventional full-reference use.

Your downloaded live stream generally has **no pristine reference**, so you cannot simply say:

"VMAF = 82, therefore the video is objectively 82/100."

Instead, for your application:

No reference

Use:

- BRISQUE/NIQE-type no-reference metrics
- blur metrics
- blockiness
- CAMBI
- noise
- exposure
- temporal stability
- effective resolution

With reference

Add:

- VMAF
- SSIM
- MS-SSIM
- PSNR
- VIF
- ADM

That distinction is important. VMAF is a perceptual quality assessment system rather than a universal standalone “video quality meter.” [GitHub +1](#)

16. The scoring system I'd actually implement in your software

I'd make the analyzer output something like:

AI VIDEO QUALITY ANALYSIS			
INPUT			
Resolution	1920*1080		
FPS	30.00		
Codec	H.264		
Bitrate	2.84 Mbps		
VISUAL ANALYSIS			
Compression	31/100	● HIGH PROBLEM	
Blur	44/100	● PROBLEM	
Temporal	87/100	● GOOD	
Texture	48/100	● LOW	
Sharpness	52/100	● SOFT	
Noise	72/100	● GOOD	
Resolution	61/100	● MODERATE	
Exposure	83/100	● GOOD	
Color	89/100	● GOOD	
Stabilization	78/100	● GOOD	
OVERALL 57.8/100			
RESTORATION PLAN			
✓ Compression restoration			
✓ Moderate deblur			
✓ Detail restoration			
X Heavy denoise			
X Color correction			
X Stabilization			
✓ Temporal protection			
AI PROCESSING LEVEL: MODERATE			

This is the architecture I'd recommend for your project: don't ask an enhancement model to blindly process every frame. First perform a relatively cheap GPU/CPU analysis, determine *which degradation actually exists*, then activate only the necessary restoration stages. That approach is especially suitable for your long 1–5 hour journalism videos because it can avoid expensive processing where the source is already good.

And one final principle: **the analyzer should score degradation, not beauty**. A naturally sharp face with normal skin texture should not be “improved” merely because an AI model can make it sharper. That is how you avoid the plastic/uncanny look you're specifically trying to avoid.